

ATPL EASA STUDY GUIDE | 050 - METEOROLOGY

Chapter 1: The Atmosphere: Structure, Composition & ISA

Syllabus Source: CAE Oxford (pp. 1-32) | Pass Target: >= 90% AviationExam

1. COMPOSITION & VERTICAL THERMAL STRUCTURE

The atmosphere is a mechanical mixture of gases surrounding the Earth, structured into concentric thermal layers:

- Gas Composition of Dry Air: Nitrogen (N2) = 78.08%, Oxygen (O2) = 20.95%, Argon (Ar) = 0.93%, Carbon Dioxide (CO2) = 0.04%. Water vapor varies from 0% (polar/desert) to 4% (humid tropics).
- Troposphere: Lowest layer containing 75% of atmospheric mass and virtually all weather (clouds, precipitation). Temperature decreases with altitude. Upper boundary: Tropopause.
- Tropopause Height Variation: Pole: ~8 km (26,000 ft) with temperature ~ -45°C to -50°C. Mid-latitudes: ~11 km (36,090 ft) at -56.5°C. Equator: ~16 to 18 km (55,000 ft) with temperature ~ -75°C to -80°C (COLDEST TROPOPAUSE IS OVER THE EQUATOR!).
- Stratosphere: From Tropopause to Stratopause (~50 km). Temperature increases with altitude due to UV absorption by Ozone (O3). Smooth air, virtually zero water vapor.

2. THE ICAO STANDARD ATMOSPHERE (ISA)

ISA provides an internationally agreed baseline for calibrating altimeters and aircraft performance:

ISA Parameter	Standard Sea Level Value	Lapse Rate / Boundary Specification
ISA Parameter	Standard Sea Level Value	Lapse Rate / Boundary Specification
Mean Sea Level Pressure (MSL)	1013.25 hPa (mbar) = 29.92 inHg = 760 mmHg	Decreases ~ 1 hPa per 27 ft (or 30 ft) at sea level.
Mean Sea Level Temperature	+15.0°C (288.15 Kelvin)	Lapses at exactly 2.0°C / 1,000 ft (0.65°C / 100 m).
Mean Sea Level Air Density	1.225 kg/m^3	Decreases with altitude and increasing temperature.
Speed of Sound at MSL (a_0)	340.3 m/s = 661.5 knots	Depends solely on absolute temperature: $a = 38.94 \times \sqrt{T_K}$.
ISA Tropopause Boundary	11,000 meters = 36,090 ft	Temperature reaches -56.5°C (216.65 K), remaining ISOTHERMAL up to 20 km (65,600 ft).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ISA Temperature & Delta ISA Calc

Problem: An airliner cruises at FL 310 (31,000 ft). Cockpit OAT reads -41 C. Step 1: Base ISA Sea Level Temperature = +15.0 C. Step 2: Calculate standard temperature decrease up to FL 310: - ISA Lapse Rate = 2.0 C per 1,000 ft. - Total drop = 31 x 2.0 C = 62.0 C. Step 3: ISA Temperature at FL 310 = +15.0 C - 62.0 C = -47.0 C. Step 4: Delta ISA = Actual OAT - ISA Temperature = -41.0 C - (-47.0 C) = +6.0 C (ISA +6 C, Warmer than standard).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating ISA Temperature & ISA Deviation

SCENARIO (Fundamental AviationExam Question): An airliner is cruising at FL 330 (33,000 ft). The cockpit Outside Air Temperature (OAT) gauge reads -45°C. QUESTION: What is the ISA temperature at FL 330, and what is the ISA Deviation (Delta ISA)? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the ISA Standard Sea Level Temperature: - ISA MSL Temperature = +15.0°C. Step 2: Calculate the standard temperature decrease up to FL 330: - ISA Lapse Rate = 2.0°C per 1,000 ft. - Total temperature decrease = $33 \times 2.0^\circ\text{C} = 66.0^\circ\text{C}$. Step 3: Calculate ISA Temperature at 33,000 ft: - ISA Temp = MSL Temp - Temperature Decrease - ISA Temp = $+15.0^\circ\text{C} - 66.0^\circ\text{C} = -51.0^\circ\text{C}$! Step 4: Calculate the ISA Deviation (Delta ISA): - Formula: Delta ISA = Actual OAT - ISA Temperature - Delta ISA = $-45.0^\circ\text{C} - (-51.0^\circ\text{C}) = -45.0^\circ\text{C} + 51.0^\circ\text{C} = +6.0^\circ\text{C}$! FINAL ANSWER: ISA Temperature at FL 330 = -51°C. ISA Deviation = ISA +6°C (WARMER than standard).

[!] EXAM TRAP (AVIATIONEXAM) - Coldest Tropopause Exam Trap

- The tropopause is HIGHEST and COLDEST over the EQUATOR (17 km / -80°C)! - It is LOWEST and WARMEST over the POLES (8 km / -45°C). - Many students incorrectly guess that the poles have the coldest tropopause. That is false!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What are the standard values for temperature and pressure in the ICAO Standard Atmosphere (ISA) at Mean Sea Level?

- [A]: 0°C and 1000 hPa
- [B]: +15°C and 1013.25 hPa
- [C]: +20°C and 1013.25 hPa
- [D]: +15°C and 1000 hPa

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The ICAO Standard Atmosphere defines MSL datum as +15°C (288.15 K) and 1013.25 hPa (29.92 inHg).

Q2: Where on Earth is the tropopause physically highest and its temperature coldest?

- [A]: Over the North and South Poles
- [B]: Over the Equator (height ~16-18 km, temperature ~ -75°C to -80°C)
- [C]: Over mid-latitude depressions
- [D]: Over the Sahara desert

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Intense convective heating expands the equatorial troposphere to 17-18 km. Because temperature lapses continuously through this deep column, the equatorial tropopause is the coldest.

Q3: Up to what altitude does the ISA temperature lapse rate of 2°C per 1,000 ft remain valid before becoming isothermal?

- [A]: 20,000 ft
- [B]: 36,090 ft (11,000 meters), where temperature reaches -56.5°C

- [C]: 50,000 ft
- [D]: 45,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ISA definition, the lapse rate terminates at 11 km (36,090 ft) at -56.5°C, where the isothermal stratosphere begins.